

Predicting Business Credit

Applications for Universal Plus: A CRISP-DM Analytics Approach

1. Literature Review

Credit risk modelling typically follows structured frameworks like CRISP-DM, which involves cycling between understanding the business problem, exploring the data, preparing it and building models (Schröer et al., 2021). Exploratory analysis is useful for spotting patterns in distributions, identifying outliers and understanding missingness before deciding on preprocessing approaches (Tripathi et al., 2021).

One of the main challenges with credit datasets is missing data. Research suggests that simple imputation methods like mean or median can work quite well when paired with robust machine learning models, especially when it is unclear why data is missing (Perez-Lebel et al., 2022). Feature selection also plays an important role in reducing noise, with filter-based methods using Information Gain commonly used in credit scoring to rank which variables are most predictive (Cai et al., 2018).

Class imbalance is another issue that comes up frequently. Rather than just resampling, recent studies point to the usefulness of adjusting class weights within the algorithm itself, which penalises errors on the minority class more heavily (Alam et al., 2020). This approach has been shown to work well in gradient-boosting models without needing to create synthetic data (Mushava and Murray, 2022).

When it comes to choosing a model, Logistic Regression is still seen as the standard for regulatory purposes because it is easy to interpret. Benchmarking studies have found that tree-based ensembles generally achieve better predictive performance on real credit datasets (Lessmann et al., 2015). Support Vector Machines (SVM) can also be accurate, but traditional non-linear SVM approaches are computationally expensive, which limits their practicality for large-scale credit scoring (Harris, 2015). More broadly, complex models like SVMs and

ensemble methods are harder to explain (Dumitrescu et al., 2022), and with regulators pushing for transparency, simpler models remain popular (Bücker et al., 2022). This highlights the ongoing trade-off between accuracy, scalability and interpretability in credit risk modelling.

2. Business Understanding

Universal Plus operates in a highly competitive business lending environment. One effective method for the bank to increase business lending revenue and relevant market share in this environment is to identify prospective business credit applicants in its current customer base, allowing the bank to proactively engage with these potential applicants, leading to more valuable business lending opportunities and improved connections with customers' respective businesses.

Using anonymised customer transaction data provided by Universal Plus, we have developed a model that identifies these potential applicants. In the following report, we will discuss our development process, with reference to the CRISP-DM framework, and how it addresses the business problem.

3. Data Understanding

The dataset contained 100,000 customer records with 200 anonymised numerical features (var_0 to var_199), an ID_code, and a binary target indicating whether the customer applied for business credit. Because all variables were anonymised, no business meaning or domain knowledge was available, so the initial focus was on understanding the structure, quality, and behaviour of the data through statistical exploration.

Summary statistics and distribution plots were used to inspect variable ranges, identify potential outliers, and understand general data shape. This revealed that all features were continuous, some contained extreme values, and missingness was minimal except for var_45, which showed approximately 40% missing entries. Examination of the target variable highlighted a severe imbalance (about 10/90), which signalled that the data preparation and modelling stages would need to account for this. These insights guided the design of the preparation steps, particularly around imputation, feature reduction, and handling the class imbalance in a way that would not distort the underlying data.

4. Data Preparation

Data preparation focused on handling outliers and missing values, reducing dimensionality, and creating training and test sets. Although several variables displayed extreme values, the dataset's fully anonymised nature meant it was impossible to determine whether these were genuine behaviours or data errors. To avoid removing potentially informative signals, no outlier removal was carried out, and extreme values were left for the modelling stage to manage.

Missing-value analysis showed that var_45 contained around 40% missing entries. To assess the impact of this, two preprocessing pipelines were created: Pipeline A removed var_45, while Pipeline B retained all features. For both pipelines, remaining missing values were imputed using the median, a simple and robust approach widely used in credit-risk modelling. This two-pipeline design aligns with CRISP-DM's iterative approach by allowing us to check whether findings were sensitive to specific preparation methods.

The dataset exhibited a strong class imbalance, so no oversampling or synthetic methods were used. Instead, embedded model balancing was applied:

- `class_weight = "balanced"` for Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM)
- `scale_pos_weight` tuning for XGBoost and LightGBM
- SVM was trained on the smaller dataset (contains a random 20% subset) due to computational constraints

To reduce dimensionality, Information Gain (IG) was calculated on the original training data, and only features with positive IG values were retained. Finally, both pipelines were split into training (80%) and test (20%) sets using stratified sampling, ensuring fair and consistent evaluation across models.

5. Modelling

This report covers six machine learning classifiers across different applications: Logistic Regression, SVM, Decision Tree, Random Forest, XGBoost, and LightGBM. Each model

follows the same data preparation method mentioned in Section 4. Tabulated below are the scores of all models for both pipelines.

Table 1: Pipeline Comparison						
Model	Pipeline	AUC	Recall	Precision	F1 Score	Accuracy
Logistic Regression	A	0.854	0.771	0.272	0.402	0.770
Logistic Regression	B	0.855	0.770	0.271	0.401	0.769
SVM (small dataset)	A	0.850	0.728	0.268	0.392	0.778
SVM (small dataset)	B	0.845	0.720	0.269	0.391	0.780
Decision Tree	A	0.620	0.632	0.140	0.230	0.574
Decision Tree	B	0.620	0.632	0.140	0.230	0.574
Random Forest	A	0.779	0.000	0.000	0.000	0.900
Random Forest	B	0.773	0.000	0.000	0.000	0.900
XGBoost	A	0.861	0.501	0.499	0.500	0.899
XGBoost	B	0.855	0.495	0.476	0.485	0.895
LightGBM	A	0.860	0.643	0.370	0.470	0.854
LightGBM	B	0.858	0.643	0.362	0.463	0.850

Comparison of preprocessing methods during the exploratory analysis showed Pipeline A outperformed Pipeline B (Table 1). Consequently, Pipeline A was chosen for model development and evaluation.

5.1 Logistic Regression (LR)

In the model, the main change from default parameters was setting max_iter=10000 to allow the solver to successfully converge and find the optimal solution. The model achieved a training accuracy of 0.77.

5.2 Support Vector Machine (SVM)

As mentioned in Section 4, we used the smaller dataset for SVM as the computational requirement for SVM model fitting is high. We experimented with both linear and non-linear (RBF) kernels for SVM and found that linear SVM performed slightly better with a training accuracy of 0.79.

5.3 Decision Tree (DT)

For Decision Tree, we first conducted a hyperparameter tuning using GridSearchCV prioritising recall score. Based on the tuning used, the best Cross Validation (CV) score of 0.56 corresponded with the best max depth of 3 (refer to Figure 1). Thereafter, the

DecisionTreeClassifier was used with criterion = 'entropy' and max depth of 3 which resulted in 8 leaves and training accuracy of 0.57.

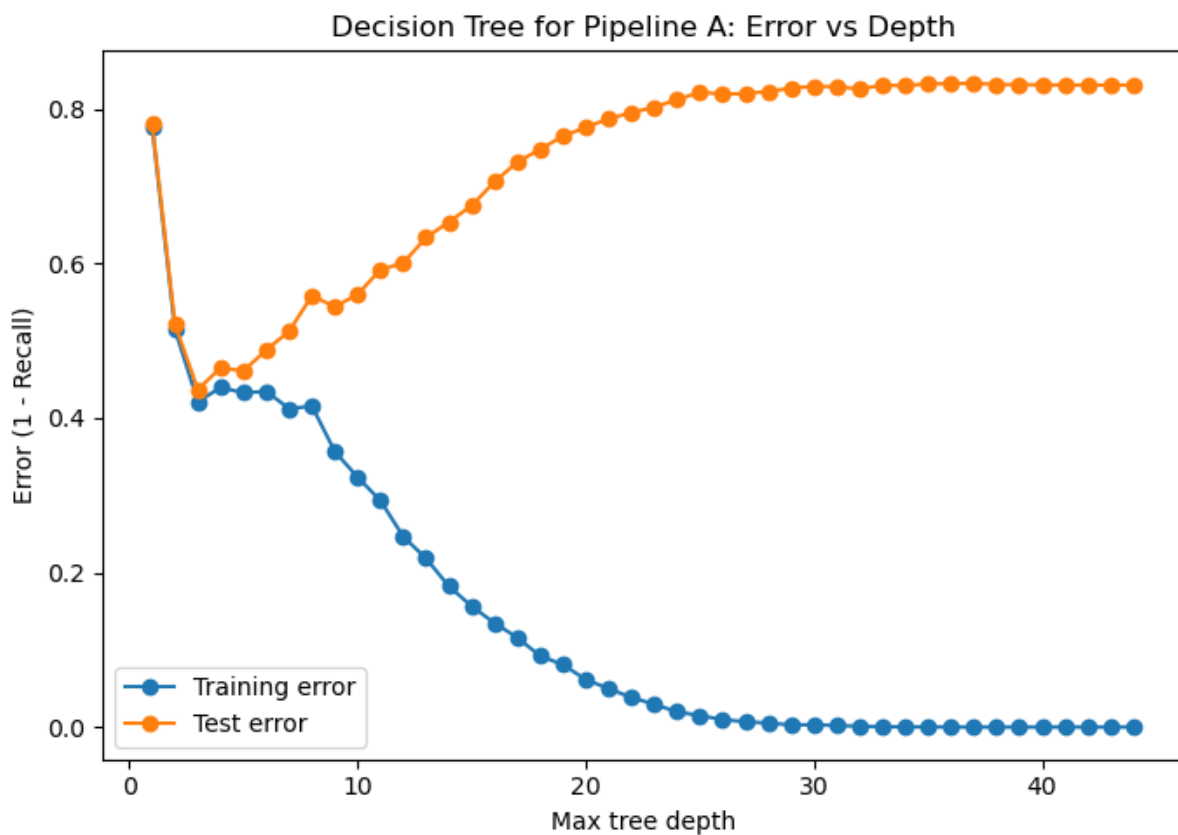


Figure 1: Optimising Recall via Grid Search Cross Validation for Decision Tree

5.4 Random Forest (RF)

A Random Forest classifier was trained using 100 trees and the square root of the total number of features at each split. The model was trained successfully and achieved a training accuracy of 1.00.

After fitting, we inspected the feature importance scores to understand which variables influenced the model most. Finally, predictions were generated on the held-out test set using the same selected features, allowing us to evaluate how well the model generalised to unseen customers.

5.5 XGBoost (XGB)

XGBoost was chosen as one of the main boosting models. The model used 300 boosting rounds with a maximum tree depth of 7, a moderate learning rate (0.05), and regularisation terms (gamma, reg_alpha, reg_lambda) to reduce overfitting. Additional parameters such as

subsample and colsample_bytree introduced randomness to improve generalisation, while the hist tree method ensured efficient training on the large dataset.

5.6 LightGBM (LGBM)

LightGBM was also used as alternative gradient boosting method. The model was trained using 300 boosting rounds with a maximum depth of 6 and 40 leaves, which offered a controlled balance between flexibility and overfitting. A moderately low learning rate of 0.03 was used so that the model learned gradually, a common recommendation in gradient boosting literature to improve generalisation performance. Regularisation terms (reg_alpha, reg_lambda and min_split_gain) were also incorporated to stabilise model behaviour and reduce sensitivity to noise.

6. Evaluation

Model performance was evaluated using five key metrics: AUC, recall, precision, F1 score, and test accuracy. These collectively assess both the ranking ability of a classifier and the balance between false positives and false negatives. For Universal Plus, recall is critical because false negatives (missing customers who later apply for business credit) mean lost revenue and weakened customer relationships. Precision and F1 score are also considered, but mainly as secondary measures, since the business can tolerate outreach to some customers who ultimately do not apply.

Table 2: Pipeline A Results Comparison					
Model	AUC	Recall	Precision	F1 Score	Accuracy
Logistic Regression	0.854	0.771	0.272	0.402	0.770
SVM (small dataset)	0.850	0.728	0.268	0.392	0.778
Decision Tree	0.620	0.632	0.140	0.230	0.574
Random Forest	0.779	0.000	0.000	0.000	0.900
XGBoost	0.861	0.501	0.499	0.500	0.899
LightGBM	0.860	0.643	0.370	0.470	0.854

Table 2 summarises the performance of all Pipeline A models. Logistic Regression emerged as the strongest model, achieving the highest recall (0.77) and maintaining a strong AUC score (0.85). Referring to Figures 2, 3, and 4, we noted that LR consistently outperformed other models, ranking it near the top of each chart. This indicates that the model correctly identifies the greatest proportion of applicants while also ranking customers reliably by likelihood.

Modelling SVM on the small dataset showed that it had a high recall (0.73) and AUC score (0.85) as well. For a fair comparison, we ran the other models using the small dataset and observed that SVM performed better across all scores except for recall. However, due to the demanding computational requirement, we concluded that SVM was not suitable for this analysis. For Decision Tree, it showed a lower value compared to other models despite tuning for recall. Additionally, optimising for recall using grid search may have negatively affected other scores, which might have better scores with increased depth. Although XGBoost and LightGBM both recorded slightly higher AUC values of 0.86, their recall values were substantially lower (0.50 and 0.64 respectively), making them less suitable in this business context where missing true positives is costly. Random Forest, despite achieving the highest test accuracy, predicted no positive cases at all and therefore returned zero precision, recall, and F1 score. This highlights why accuracy alone is misleading in heavily imbalanced datasets.

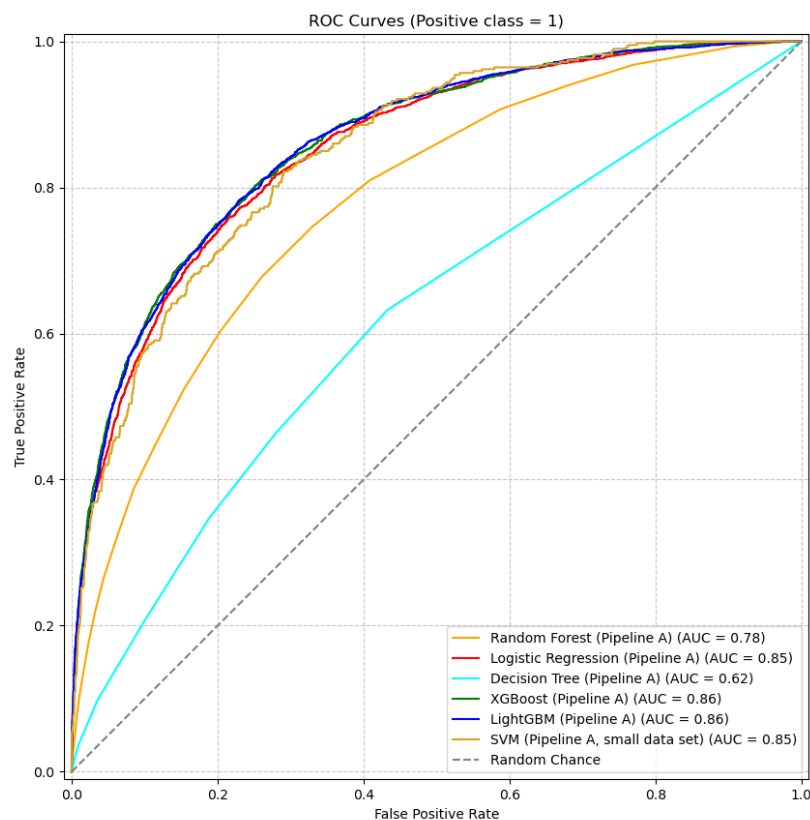


Figure 2: ROC Comparison between All Models

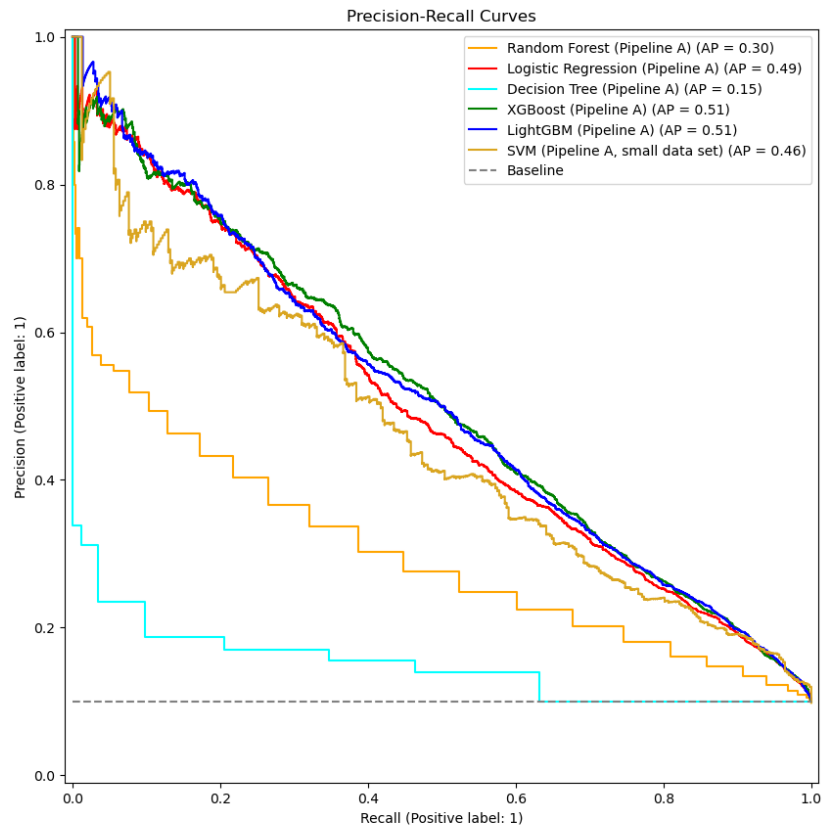


Figure 3: Precision-Recall Comparison between All Models

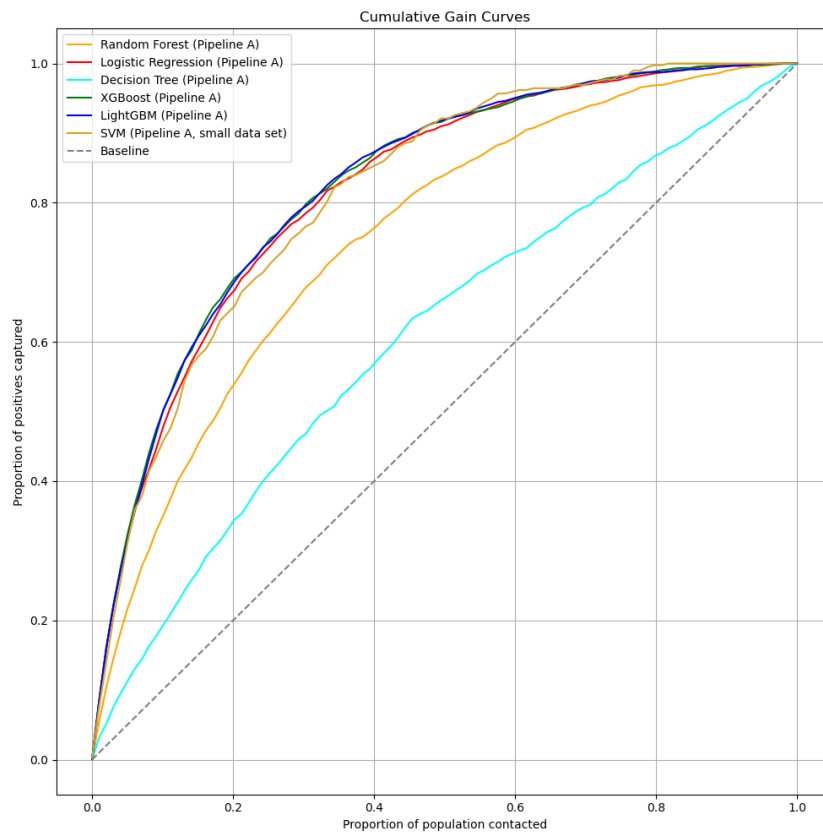


Figure 4: Cumulative Gain Comparison between All Models

After selecting Logistic Regression as the optimal model, we assessed its detailed performance to ensure that its behaviour aligned with the business objective. The confusion matrix (Figure 5) shows 1,549 true positives and 459 false negatives, confirming that the model successfully captures a meaningful proportion of likely applicants. This supports its use as a prioritisation tool for customer engagement teams. Furthermore, permutation importance analysis enabled us to rank the top contributing variables (Figure 6), providing interpretability that is valuable for regulatory explainability and internal stakeholder trust.

Overall, the evaluation demonstrates that Logistic Regression delivers the best balance between predictive performance, interpretability, and business impact, making it the most appropriate model for deployment within Universal Plus's customer-engagement processes.

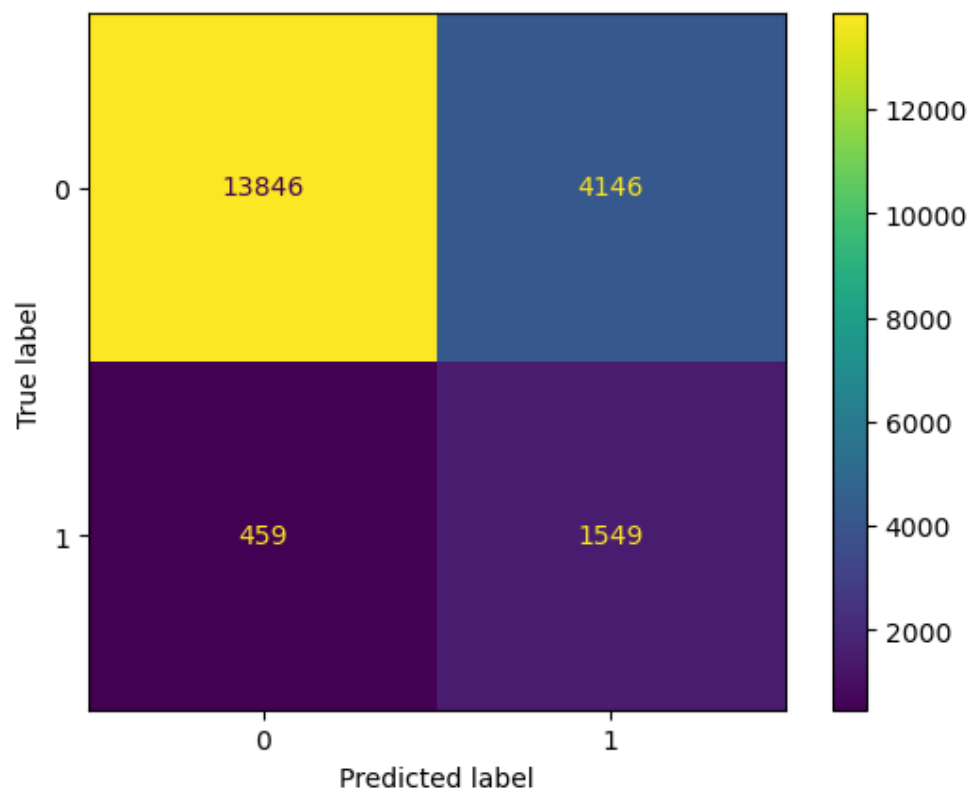
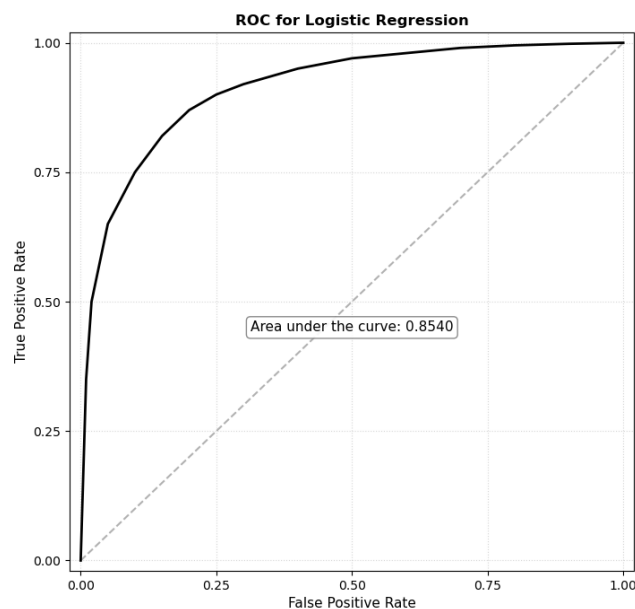


Figure 5: Confusion Matrix - Logistic Regression (Pipeline A)

Model	Accuracy	Recall	F1 Score	AUC
Logistic Regression	0.77000	0.77100	0.40200	0.85400



Top 10 Independent Variables	
Variables	Importance
var_81	100.0
var_139	70.3
var_12	70.0
var_53	65.9
var_22	64.1
var_6	58.0
var_26	54.1
var_146	52.9
var_76	52.8
var_99	52.3

Figure 6: Logistic Regression Model Evaluation Results

7. Deployment

The deployment stage focuses on how the final Logistic Regression model can be used operationally within Universal Plus to support proactive customer engagement.

Implementation would begin by integrating the model into existing workflows, embedding predictions and recommendations into relationship manager (RM) dashboards within the CRM environment. A pilot phase would then test the model with select branches or customer segments to validate results before wider rollout. Training and adoption efforts would equip RMs to interpret the model's insights and take proactive actions, such as prioritising outreach to customers with elevated credit-application likelihood. Continuous monitoring would track model performance, adoption rates and business outcomes to ensure reliability over time.

Next steps would involve scaling the model broadly across all branches and product lines once the pilot proves successful. The model would be refined and improved by incorporating feedback and new customer data to enhance accuracy, including periodic recalibration and

scheduled retraining. Finally, measuring impact would involve monitoring key metrics such as customer retention, cross-sell success and revenue growth to validate the model's continued business value.

8. Conclusion

This project addressed key challenges including high dimensionality, missing values and class imbalance to develop a model that identifies potential business credit applicants. Logistic Regression was selected for its strong recall (0.77), competitive AUC (0.85), and interpretability which is critical for regulatory compliance and stakeholder trust. A key limitation is the anonymised nature of the variables, which makes it hard to apply business knowledge or explain the results. With access to actual variable definitions, further refinement could improve both predictive performance and actionable insight for Universal Plus's customer engagement strategy.

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